the actual time point. We observed decreased temporal inference performance in the latter scenario (Supplemental Fig. S7A,C,E). Similarly, removing the condition factor decreases the performance of temporal inference (Supplemental Fig. S7B,D,F). When removing the batch factor from the model, we observed a misalignment of cell identity factors across different data sets, compared with well-aligned cell identity factors from the original Sunbear model (Supplemental Fig. S8). These results suggest the importance of explicitly encoding condition and batch factors and representing time with a sinusoidal encoding.

Sunbear reveals sexual dimorphism in gene expression during mouse embryonic development

Having established Sunbear's ability to capture sex differences in gene expression, we used the model to investigate sex-specific transcriptional differences during mouse embryonic development. Because only one sex is profiled at most time points, it is impossible to directly perform differential expression analysis between sexes at individual time points. Therefore, for each cell, we swapped the sex factor and calculated differential gene expression between sexes (see the Methods subsection "Predicting transcriptional sex differences across time").

We performed two types of validation of the transcriptional sex-difference predictions, taking advantage of the three time points (E13, E13.25, E13.75) for which the original data include both female and male samples. Specifically, we performed evaluation on each cell type that has more than 1000 cells profiled across the three time points. First, we compared our prediction against the differential expression pattern between sexes calculated based on the original data at each sex-matched time point. For each of the three time points, we used the area under the receiver operating characteristic curve (AUROC) to assess the extent to which Sunbear's sex-difference prediction identifies the female- and male-biased genes derived from the original data set. As a baseline, we used the differential expression pattern from the closest time point as the sex-difference prediction for the time point of interest (see the Methods subsection "Validation of sex-difference prediction"). Comparing the two, we found that Sunbear significantly outperforms the baseline method (one-sided Wilcoxon paired rank-sum tests $P = 1.24 \times 10^{-5}$) (Fig. 3A). In the second validation, we compared both the predicted and original sex difference patterns against a small set of genes known to escape X Chromosome inactivation (XCI) in most/all mouse tissues (known constitutive escape genes; see the Methods subsection "Validation of sex-difference prediction") (Berletch et al. 2015). Escape genes are transcribed from both copies of the X Chromosome in females (XX) but only from the one X copy in males (XY); thus, they are expected to have higher expression in females than males. Sunbear is able to rank the known escape genes to be more highly female-biased than other genes on the X Chromosome (Fig. 3B). In contrast, these genes are not ranked highly when we perform differential expression analysis directly on the original data (one-sided Wilcoxon paired rank-sum tests $P = 8.36 \times 10^{-11}$) (Fig. 3B). This trend is consistent when comparing based on area under the precision-recall curve (Supplemental Fig. S9). These results suggest that Sunbear is able to capture time-specific and sex-biased transcription patterns.

We then systematically investigated the sex-biased pattern of two major cell trajectories across all time points: macrophages and central nervous system (CNS) neurons (see the Methods subsection "Validation of sex-difference prediction"). Specifically, we focused on the most abundant cell types within each trajectory: glutamatergic neurons and border-associated macrophages. Macrophages have been widely studied in the adult stage and exhibit well-known sex-biased transcription and phenotypes (Gal-Oz

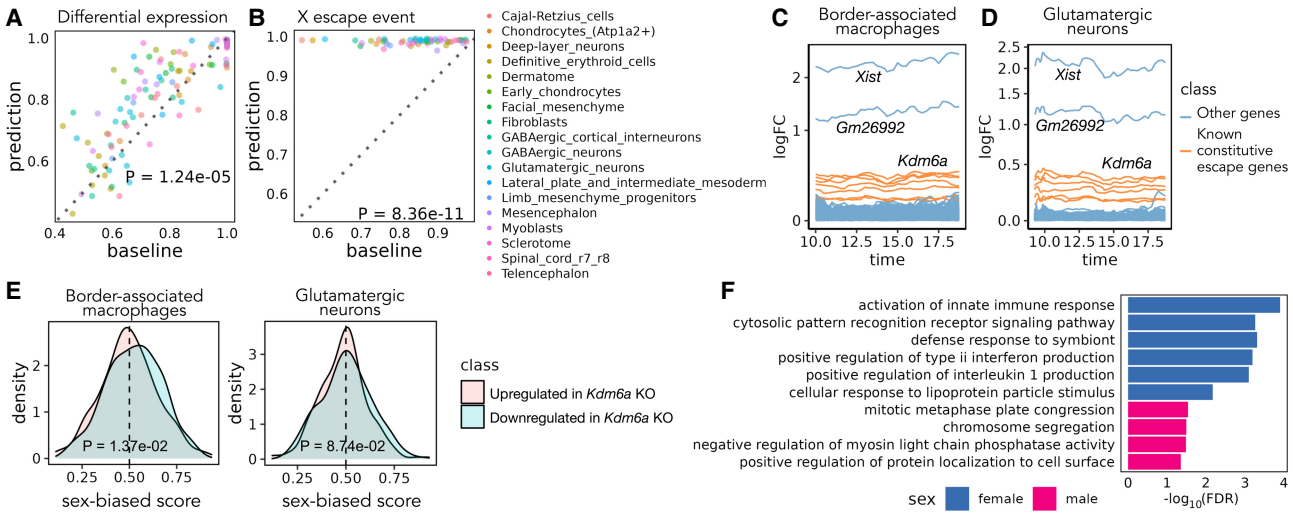

Figure 3. Sex differences in mouse embryonic development. (*A*) Pairwise comparison of Sunbear prediction and the nearest neighbor baseline in recapitulating differential expression patterns in sex-matched time points. Each dot indicates the AUROC score of recapitulating female-/male-biased patterns in each sex-matched time point and cell type. (*B*) Similar to in *A*, pairwise comparison of Sunbear prediction and the nearest neighbor baseline in ranking escape genes to be more female-biased than all other genes on the X Chromosome. (*C*,*D*) Predicted temporal sex-biased log fold change in glutamatergic neurons and border-associated macrophages. Each line represents a gene that is predicted to be consistently higher in females than males and is colored by whether the gene is a known constitutive escape gene or not. (*E*) Distribution of predicted sex-biased scores of genes (0 = extremely male-biased, 1 = extremely female-biased), grouped and colored by whether the gene is upregulated (pink) or downregulated (blue) in *Kdm6a* KO versus WT samples in $CD4^+$ cells. *P*-values are calculated by one-sided Wilcoxon rank-sum tests. (*F*) Gene Ontology biological processes enriched in consistently female- and male-biased genes in border-associated macrophages. Nonredundant terms with the smallest FDR are selected for visualization. No enrichment of biological processes is found in glutamatergic neurons.